Andrew Crossland

PhD MEng MSc MIET CEng FEI

https://www.linkedin.com/in/afcrossland/,
www.mygridgb.co.uk

Introduction

I am a passionate energy systems expert who has delivered innovate new business models, led corporate strategy and created numerous world first projects. I have a broad, thorough and respected understanding of the energy transition backed by strong communication skills, a strategic approach, deep modelling expertise, thought leadership and teamworking.

I have track record of creating and leading to bring innovative and first of kind new energy projects around the world– from standards to whole island electrification.

Notable Achievements and Skills

- Interdisciplinary assessment of energy systems – financial, technical, social factors from my PhD through to commercial work. Key to my ability to maximise value from new energy projects.
- Director level experience. Advisor at Board and CEO level to numerous startups inc. retailers and low carbon heat.
- A long term and active role with Durham University and Durham Energy Institute through lectures, supporting researchers, independent research, as Advisory Board Member of the DEI and now Visiting Professor in Practice.
- Global knowledge – experience in 20+ countries and across four continents.
- Directed consultancies Advance Further Energy Ltd and Future Zero Consulting Ltd.
- Track record at political level including Parliamentary presentations (UK/NZ), advising Ministers, Chairing National Working Groups and delivering regulatory change.
- Developed first utility scale battery storage system in New Zealand.
- Fellow of the Energy Institute by the “Eminence Route” in June 2020.
- Author of “Decarbonising Electricity: Made Simple”, a book which shows how to achieve low carbon transitions.
- 27,000+ followers on social media tracking British Electricity Transition on MyGridGB. Associated energy balancing model and digital twin of 2030 grid. MyGridGB energy transition podcast with 2,000+ listeners.
- Experienced communicator and public speaker. Have worked for a wide variety of audiences from public sector, academia, and industry – including two national Parliaments!
- Software including Dataiku, OpenDSS. AutoCAD, PSCAD, Simulink, PVSyst, HOMER, MATLAB, RTDS.
- Energy UK “Rising Star” in 2018. Subsequently nominated for two further awards by Energy Institute.

Present Advisory Roles

Jan 2022 - Advisor, Powering Africa Recharging Energy

- Developing innovative commercial model for delivering high power microgrids across East Africa.
- Technical and commercial support, innovation and capacity building.
- Supporting multi-million-dollar investments in decentralised energy systems to support conservation.

Oct 2016 - Associate Professor in Practice, Durham Energy Institute and School of Engineering, UK

- Voluntary role supporting education, research and most importantly, people.
- Sitting on advisory board of global Durham Energy Institute.
- Delivering annual set piece lecture to students on careers in energy.

Previous Roles

May 2023 – Nov 2024 Director of New Energy, SolarZero, Global

- Leading global growth strategy for SolarZero – quantifying new markets, developing strategy, advising board/exec, driving key partnerships and building teams.
- Advancing of SolarZero virtual power plant: connecting 13,000 batteries across New Zealand and providing services to consumers and to the grid. Data science architect and delivery lead.
- Creating new batteries and heat pumps with major Japanese manufacturer

Apr 2022- May 2023 Head of Battery Storage, Renewco Power Ltd, Global

- Managed and developed corporate strategy for energy storage, hydrogen and flexibility.
- 500MW+ battery energy storage portfolio in Midwest USA and UK within 6 months.
- Delivery of development tool for grid and land assessments (battery/wind/solar/hydrogen)

May 2018 – Apr 2022 Energy Strategy and Development Specialist, Infratec, New Zealand

- Created development process with 200MW pipeline in 12 months.
- Lead bid manager on $50m+ pipeline of microgrids projects including 4x microgrids in The Solomon Islands, 7x islands in Tonga, and major projects in Tuvalu.
- Consultancy and pilots with National Grid in New Zealand leading to first of a kind storage pilots including first frequency response from a battery in New Zealand.
- Multi-disciplinary modelling of energy systems including solar, traditional generators and storage.
- Policy support including presentations to Minister of Energy of New Zealand.
- Capacity building with utilities, Government, academia using lectures, research and consultancy.

Sep 2018 – May 2022 Director and Lead Consultant, Advance Further Energy Ltd, UK

- Advised energy start-ups in battery storage, energy trading and demand response.
- Delivered MCS Guidance Note MGD-003 and associate solar/energy storage standard – applied on every domestic solar quotation across the UK.
- Construction and validation of bespoke energy modelling tools for emerging applications.

Dec 2015 -May 2018 Energy Storage Specialist, SolarCentury, UK

- Energy system modelling – technical and commercial. IPP/PPA projects. Turnkey.
- Full lifetime technical and commercial assessment for clients and strategic decision making.
- Delivered first fully decarbonised store in the UK. Led first retail offers for storage and solar PV with IKEA/Nissan. Largest batteries in East Africa – Kenya and Eritrea.
- Regulatory change with Government to support energy storage rollout
- Chairperson of Behind the Meter Storage Group at Solar Trade Association.
- PR exercises including drafting articles for industry and public magazines.
- Creation of bankable modelling e.g., IPP/PPA, Turnkey, whole life energy system models etc.

PhD in technoeconomic modelling of energy storage, Durham University (Oct 2011 - Sep 2014)

- Technical and financial study into renewable integration on 9,000 LV networks. Business cases for energy storage
- Quantified DNO network reinforcement savings ~£90 million per UK DNO using energy storage/heat pumps.
- The first full scale model of 2 million homes and 9,500 networks in NW England. Using that model to perform a technical and financial assessment of the impact of solar PV and storage.
- New and published procedure for extracting LV network models from GIS systems.
- New published methods to determine if and where a network operator would locate energy storage in LV networks.
- Strong working relationship with DNO throughout research. Focus on producing business relevant results.

Higher Education (other)

2005 – 2009 MEng, Durham University, UK
General Engineering with New and Renewable Energy MEng (Hons), First class. 74% average.

2009 - 2010 MSc/MBA, University of Southampton, UK
Transport and the Environment MSc, MBA modules, PhD Research. 72% average.

2015-2016 PgDip in Railway Infrastructure Engineering, Sheffield Hallam University, UK